Computer Games Development CW208

GDD

Year IV

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Acknowledgements

I would like to thank my lecturers, supervisor, and peers at SETU for their guidance throughout this project.

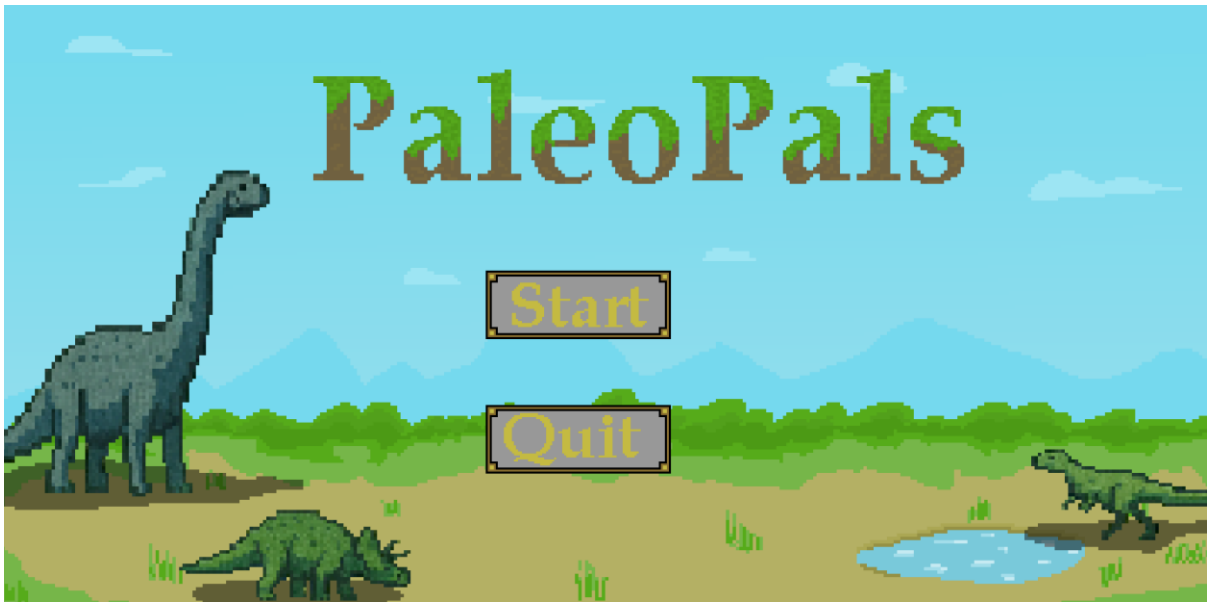
I also acknowledge the creators of SFML and the nlohmann JSON library, whose tools formed the foundation of this project.

Game Overview

PaleoPals is a 2D mining, exploration, and fossil-collection game where the player digs through layered terrain to uncover dinosaur fossils, amber, and other collectibles. These items can be displayed in a museum, sold for money, or used to unlock cosmetic skins.

The game takes place in a large, destructible underground world. The player controls a single character who mines tiles, collects items, upgrades tools, and interacts with buildings such as the Museum and Trader. As progression increases, the player can hire palaeontologists who assist with mining and fossil retrieval.

Why create this game? To explore procedural terrain, a data-driven content pipeline, and a satisfying progression loop inspired by mining and idle-automation games.



Main Focus

- Mining and exploration
- Collecting fossils and completing dinosaur skeletons
- Upgrading tools and hiring helpers
- Building a sense of discovery and progression through a JSON-driven content system

What is Different

- A fully destructible world with layered hardness - all layer data defined in JSON
- A museum interior that visually updates as fossils are collected, populated entirely from JSON species data
- A data-driven fossil system using JSON configuration - new dinosaurs and collectibles require no code changes
- A Trader upgrade system integrated with an in-game economy

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  "layers": [
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    ]
  },
```

Feature Set

General Features

- Large destructible tile-based world
- Multiple terrain layers with different hardness (JSON-configured)
- Fossils, amber, and trash collectibles (JSON-defined types and values)
- Museum interior with dynamic fossil displays (populated from JSON data)
- Trader building with upgrades and hiring
- Palaeontologist helpers
- Smooth camera system with clamping
- JSON-driven configuration for all game content

Gameplay

- Mining tiles to uncover fossils
- Collecting items for money or museum completion
- Upgrading pickaxe radius, damage, pickup radius, and jump height via the Trader
- Hiring palaeontologists to assist with mining
- Exploring deeper layers with increasing hardness
- Viewing completed dinosaurs in the Museum Interior

The Game World

Overview

The world of PaleoPals is a vertical slice of earth, with the surface at the top and increasingly dense geological layers below. The player begins at the surface, where the Museum and Trader buildings are located, and digs downward to uncover fossils.

World Feature #1 - Destructible Terrain

The entire underground world is made of destructible tiles. Each tile has:

- A terrain type (Topsoil, Sediment, Rock, Bedrock) - loaded from JSON
- Hardness and HP values - defined in JSON
- A crack overlay that updates as the tile takes damage

World Feature #2 - Fossil Ecosystem

Fossils are distributed throughout the underground layers. When tiles break, they may spawn:

- Fossil pieces (assigned to randomly selected dinosaur species from JSON data)
- Amber (small or large - value defined in JSON)
- Trash items

These are managed by the FossilManager, which loads all dinosaur and collectible data from JSON at startup.

The Physical World

Key Locations

- Surface: Museum, Trader, starting area
- Underground: 100 rows of tiles, each row representing deeper geological layers

Scale

- Tile size: 24x24 pixels
- World width: 75 tiles
- World height: 100 tiles

Objects

Tiles, Fossils, Amber, Trash, Paleontologists, Museum interior assets, Trader menu UI

Weather / Day-Night

No weather system or day/night cycle is implemented.

Time

Time is continuous and real-time. A typical three-minute gameplay slice:

- Minute 1: Player mines surface layers, finds first fossils, visits museum to view collection
- Minute 2: Player digs deeper, encounters harder tiles, collects amber for money
- Minute 3: Player returns to surface, buys upgrades or hires paleontologist

Rendering System

Overview

Rendering is handled through SFML's 2D graphics API. The game uses sprites, textures, and views to draw the world efficiently.

- All assets are 2D sprites
- Tiles are drawn with frustum culling
- Collectibles are drawn only when in view
- UI is drawn in screen space

Camera

The camera follows the player smoothly and clamps to world bounds. When menus (Trader, Museum) are open, the camera switches to the default view for UI clarity.

Game Engine

Overview

The engine is a custom C++ framework built on SFML 3. It manages game states, rendering, input, physics, tile updates, upgrade application.

The engine updates all systems in a fixed timestep loop to ensure consistent physics and mining behaviour.

Collision Detection

Collision is tile-based. The player checks tile hardness and HP to determine if mining is possible. Movement uses simple AABB checks. No water simulation is included.

The World Layout

Overview

The world is a vertical grid of tiles with increasing hardness. All layer properties are loaded from JSON.

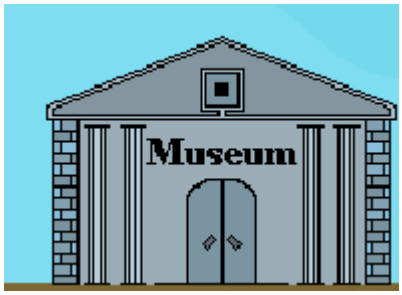
World Layout Detail #1

Topsoil -- Sediment -- Rock -- Bedrock. Each layer has unique textures and hardness values defined in config.json.

```
{
  "map": {
    "layers": [
      {"name": "Topsoil", "texture": "ASSETS/IMAGES/Terrain/Topsoil.png", "hardness": 5},
      {"name": "Sediment", "texture": "ASSETS/IMAGES/Terrain/Sediment.png", "hardness": 30},
      {"name": "Rock", "texture": "ASSETS/IMAGES/Terrain/Rock.png", "hardness": 200},
      {"name": "Bedrock", "texture": "ASSETS/IMAGES/Terrain/Bedrock.png", "hardness": 500}
    ]
  },
```

World Layout Detail #2

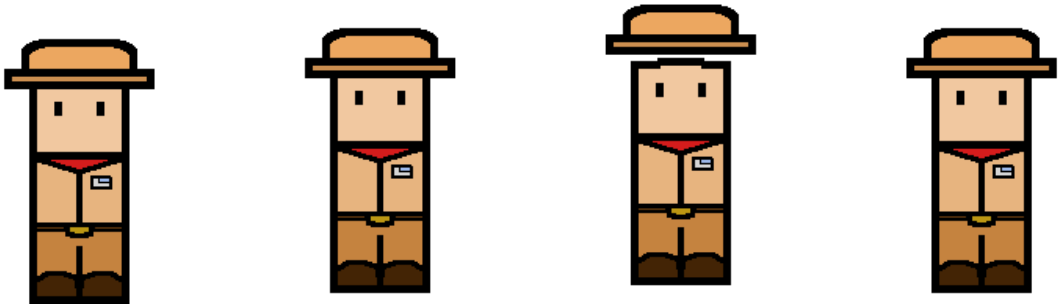
Buildings are placed on the surface using positions read from JSON. They do not move during gameplay.



Game Characters

Overview

Characters include the player and palaeontologists. Character design was inspired by safari outfits, stereotypical palaeontologist attire, and outfits from Jurassic Park.



Enemies and Monsters

There are no enemies in PaleoPals.

User Interface

Overview

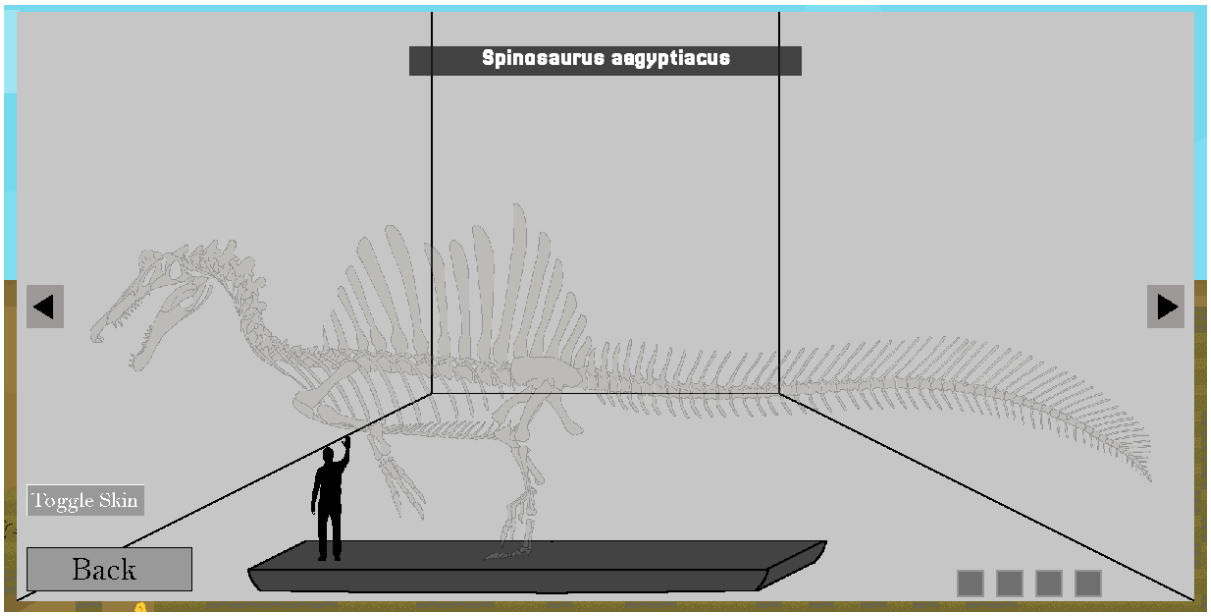
The UI includes Money display (HUD) , Trader menu , Museum interior , Pause menu ,Main menu.

User Interface Detail #1 - Trader Menu

The Trader menu uses a tabbed layout with Upgrades and Hiring tabs. Each upgrade button displays the current level, the stat effect, and the cost of the next purchase. Costs are calculated dynamically and update after each purchase.

User Interface Detail #2 - Museum Interior

The Museum Interior displays collected fossil pieces for each dinosaur species. Species data (names, piece textures, skin textures) is loaded entirely from JSON. Players can navigate between species using arrow buttons and toggle between skeleton and skin views.



Tools

The player uses a shovel to mine tiles. The pickaxe has four upgradeable stats accessible through the Trader: mining radius, damage, pickup radius, and jump height.

Mu

sical Scores and Sound Effects

Sound is minimal and not a core focus of this project. Plans existed to include free sound bites for mining, finding collectibles, completing a skeleton, and digging, but these were not implemented in the final version.

Single-Player Game

Overview

PaleoPals is a single-player mining and collection game focused on progression and discovery. There is no narrative storyline; the focus is on exploration, collection, and upgrade progression.

Single Player Detail #1

The player mines tiles, collects fossils and amber, and upgrades their tool stats at the Trader. Money earned from collectibles (values defined in JSON) funds these upgrades.

Single Player Detail #2

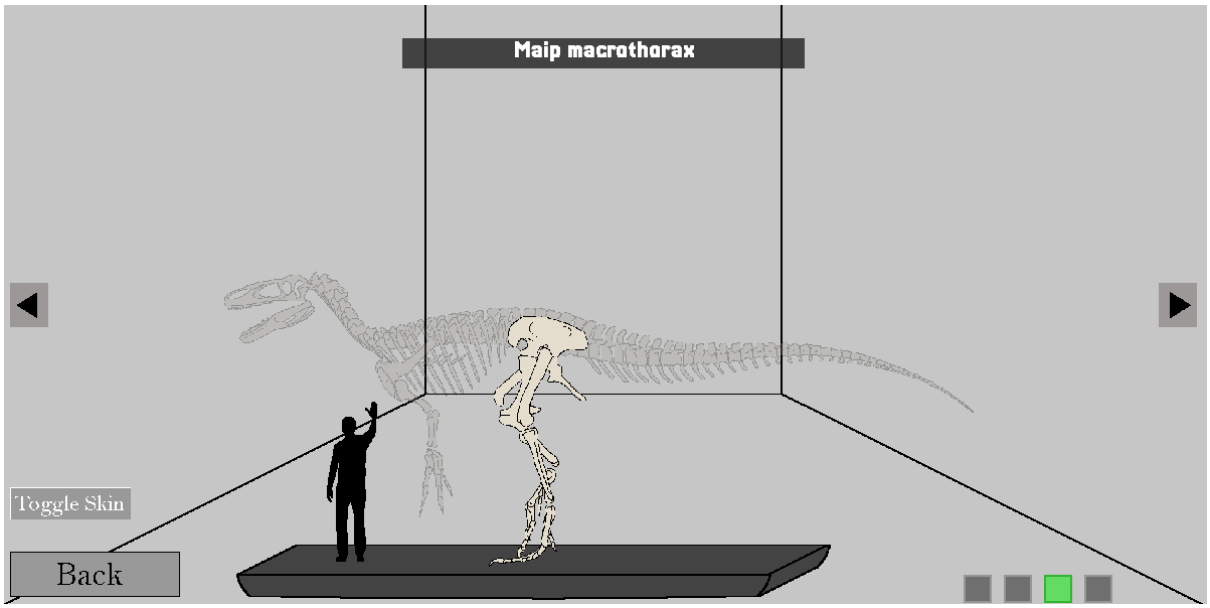
Hiring palaeontologists provides additional mining assistance and speeds up progression.

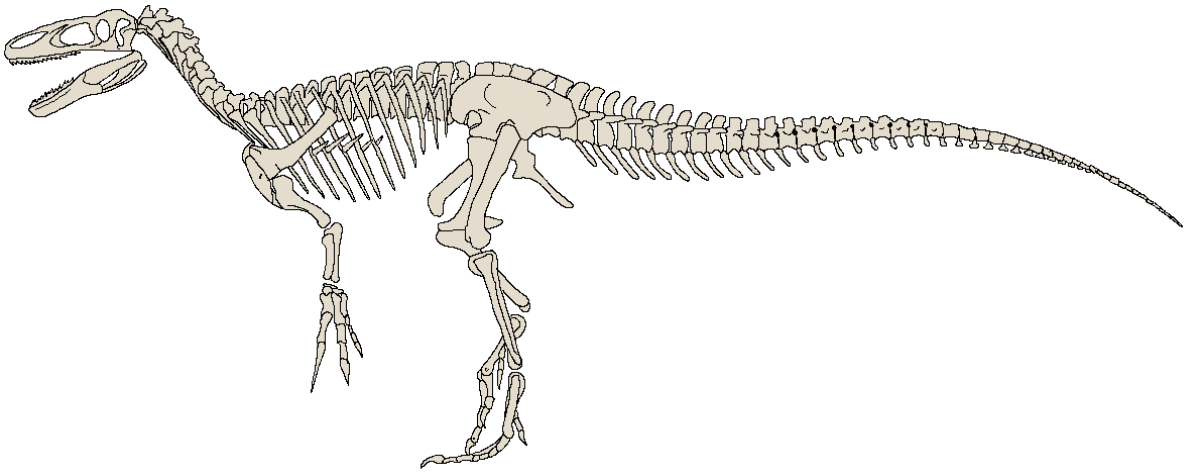
Hours of Gameplay

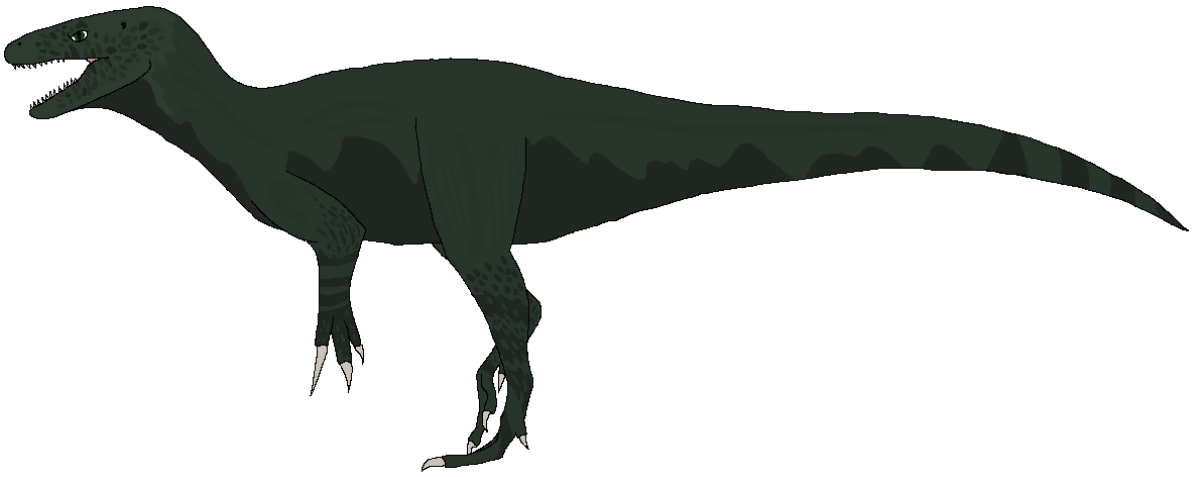
Estimated 2-4 hours for full museum completion depending on fossil RNG and upgrade choices.

Victory Conditions

The game is considered complete when all dinosaur skeletons in the museum are fully assembled.







Character Rendering

Overview

Characters are rendered as animated sprites using sprite sheets with multiple frames. Animations include idle, walking, and mining states.

World Editing

Overview

The world is generated from JSON configuration and is not edited in-game. Developers can modify config.json to change terrain layers, hardness values, building positions, collectible types, and dinosaur species. All tile hardness values and textures are fully data-driven.

References

Videos Used

Ögren, P. (2021). Behavior trees playlist [YouTube playlist]. YouTube. https://www.youtube.com/playlist?list=PLFQdM4LOGDr_vYJuo8YTRcmv3FrwczdKg

Computerphile. (2014, December 18). A search algorithm* [Video]. YouTube. https://www.youtube.com/watch?v=ySN5Wnu88nE

Natsu Games. (2023). A pathfinding explained simply* [Video]. YouTube. https://www.youtube.com/watch?v=pXsPj6HewhM

Shah, M. (2020). C++ programming playlist [YouTube playlist]. YouTube. https://www.youtube.com/playlist?list=PLvv0ScY6vfd8j-tlhYVPYgiIyXduu6m-L

Books Used

Meyers, S. (1996). More effective C++: 35 new ways to improve your programs and designs. Addison-Wesley.

Documentation Used

SFML Team. (2024). SFML 3.0.0 documentation. SFML. https://www.sfml-dev.org/documentation/3.0.0/

Websites Used

Nlohmann. (2024). basic_json API documentation. JSON for Modern C++. https://json.nlohmann.me/api/basic_json/

Applications Used

Visual Studio Code | Visual Studio 2022 | Aseprite